CCY2001: Introduction to Cybersecurity

Week 8 — Network Security Basics

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CCIT

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Outline

What is a Network?

Definition

A **network** is a collection of interconnected devices (computers, servers, phones) that communicate to share *data*, *applications*, and *resources*.

- Devices may be on different **network segments** or geographically distant
- All form a single community of workstations, servers, switches, and SaaS gateways
- Linked assets serve core **business functions**

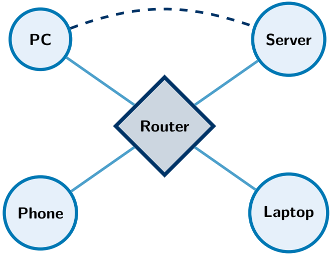
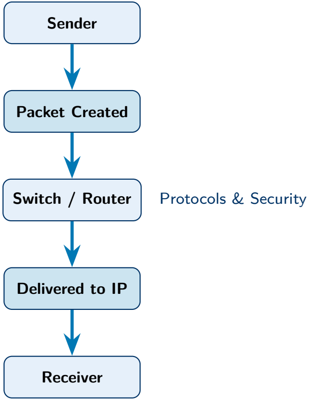


Fig. Simple LAN topology

How Does a Network Work?

- Devices (**nodes**) connect via *wired* (Ethernet, fibre) or *wireless* (Wi-Fi, Bluetooth) media
- Data travels as **packets** routed by switches and routers
- **Protocols** (TCP/IP, HTTP) govern formatting, transmission, and receipt
- Every device has a unique **IP address** for delivery
- **Security mechanisms** protect against unauthorised access

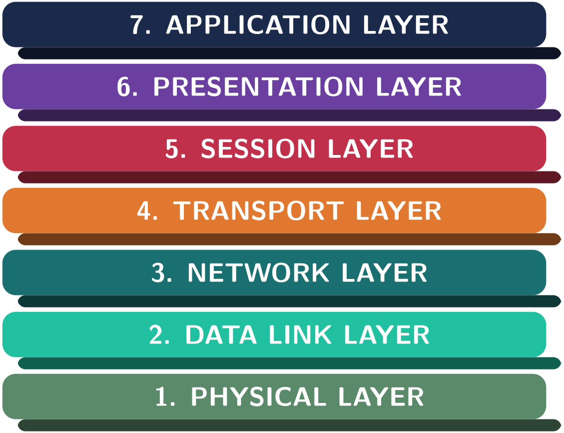


The OSI Model — Overview

What is it?

The **Open Systems Interconnection (OSI)** model describes **7 layers** that define how computer systems communicate over a network.

- Each layer has **specific responsibilities**
- Ranges from physical hardware *(Layer 1)* to application-level interactions *(Layer 7)*
- Provides a **universal language** for network communication



The OSI Model

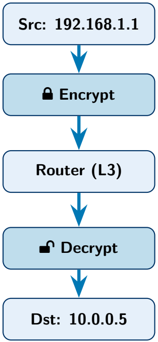
- **7 APPLICATION** → Accesses protocols used by applications to communicate data to end users.
- **6 PRESENTATION** → Configures data into an acceptable format via translation, encryption, and compression.
- **5 SESSION** → Establishes, manages, and terminates sessions between communicating devices.
- **4 TRANSPORT** → Handles data transmission between devices using TCP & UDP protocols.
- **3 NETWORK** → Conducts routing between networks, determining the most efficient path.
- **2 DATA LINK** → Handles node-to-node data transfer; assigns local addresses (MAC method).
- **1 PHYSICAL** → Converts data into bits; encodes signals, cabling, connectors, and physical specs.

Layer 3 — The Network Layer

Key Role

The **Network Layer (Layer 3)** is where data is *addressed*, *routed*, and *transmitted* across interconnected networks.

- Servers create **packets** and forward them via routing devices
- Devices are identified by **numerical IP addresses**
- **Encryption** hides packet contents from eavesdroppers
- Packets are reassembled and decoded at the destination

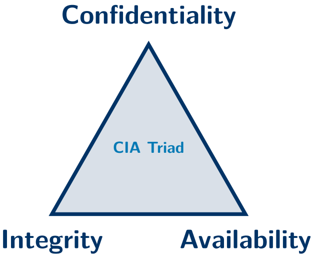


What is Network Security?

Definition

Network security encompasses the technologies, policies, people, and procedures that protect all network resources from cyberattacks, unauthorised access, and data loss — upholding the **CIA Triad**.

- Protects **hardware**, **data**, and other digital assets
- Practices can extend to **human** (user) behaviour
- A vital subset of the broader **Cybersecurity** field



Why Does Network Security Matter?

The Stakes are High

- **328.77 million terabytes** of data are created daily
- Global average cost of a data breach in 2025: **USD 4.44 million**
- Without security, organisations face **attacks, hacks, and breaches**

Network Security Enables Businesses to...

- Protect **corporate data**
- Control **network access** and availability
- **Detect and prevent** intrusions
- **Respond** to and address incidents
- Protect hardware, software, and **IP**
- Secure **data centres** and cloud resources

How Network Security Works

Defence-in-Depth Approach

A **multi-layered** strategy: authorised users gain access; malicious actors are blocked at every level.

Physical

Technical

Admin

Physical

Control physical access to routers, switches, servers

Technical

Protect data at rest & in transit via encryption

Administrative

Policies for user behaviour, authentication, authorisation

What is a Network Security Model?

Definition

A **network security model** is a conceptual framework outlining the policies, architecture, and measures designed to protect a network and its resources.

- Defines *how* security is implemented and enforced across all network layers
- Ensures the **CIA Triad**: Confidentiality, Integrity, Availability
- Establishes guidelines for **access management**, **data handling**, and **incident response**
- Goal: a secure environment defending against an **ever-evolving** cyber threat landscape

Network Security Models — Overview

Model	Core Idea	Analogy
Perimeter Security	Defends outer boundary (firewalls); trusts everything inside	*Castle with tall walls — free movement inside*
Zero Trust	"*Never trust, always verify*" — strict identity checks at every step	*Hotel keycard: inside the building $\neq$ access everywhere*
Defence-in-Depth	Multiple overlapping security layers; one failure $\neq$ total breach	*Bank: vault + guards + cameras + alarms*
Formal Models	Mathematical frameworks (Bell-LaPadula, Biba) for high-security environments	*Blueprints for a provably secure system*

Zero Trust vs. Perimeter — A Closer Look

Perimeter Security

- Traditional "*castle-and-moat*" model
- Heavy reliance on **firewalls** at the edge
- Implicit trust for *inside* traffic
- Risk: An attacker who gets inside moves freely

Zero Trust

- Modern "*never trust, always verify*" model
- **Strict identity** verification every time
- **Least-privilege** access regardless of origin
- **Micro-segmentation** limits lateral movement

Perimeter: Trust Inside $\xrightarrow{\text{Evolution}}$ Zero Trust: Verify Always

Key Network Security Technologies (1/2)

Firewalls

- Barrier between trusted & untrusted networks
- Monitor traffic based on **security rules**
- Next-gen firewalls (NGFWs) offer **deep-packet inspection** and intrusion prevention

Antivirus & Anti-malware

- Detect & remove malicious software
- Covers viruses, **ransomware**, spyware
- Deployed on network **endpoints**

Intrusion Prevention Systems (IPS)

- Scans traffic for **suspicious patterns**
- **Automatically blocks** detected threats
- Works inline in real-time

Virtual Private Networks (VPN)

- **Encrypted tunnel** over a public network
- Remote users access internal resources **safely**
- Protects data confidentiality in transit

Key Network Security Technologies (2/2)

Network Access Control (NAC)

- Authenticates **users and devices**
- Enforces policies — only **compliant** endpoints connect
- Reduces insider-threat risk

Network Segmentation

- Divides network into **isolated sub-networks**
- Limits **lateral movement** of attackers
- Contains breaches to a **smaller area**

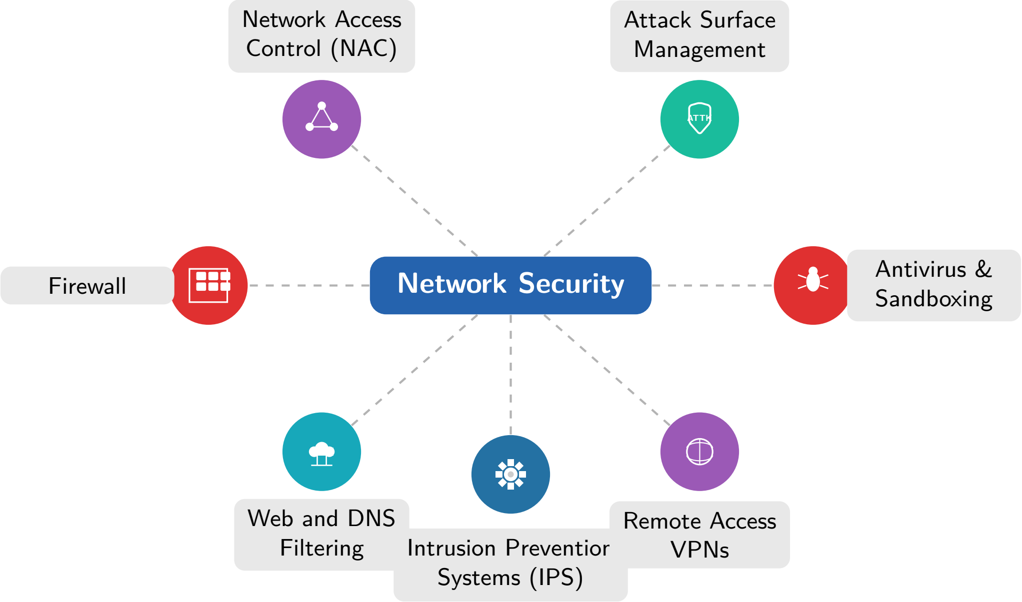
Email Security

- Blocks **phishing**, malware, and spam
- Email remains the **#1 attack entry point**
- Combines filtering, sandboxing, and training

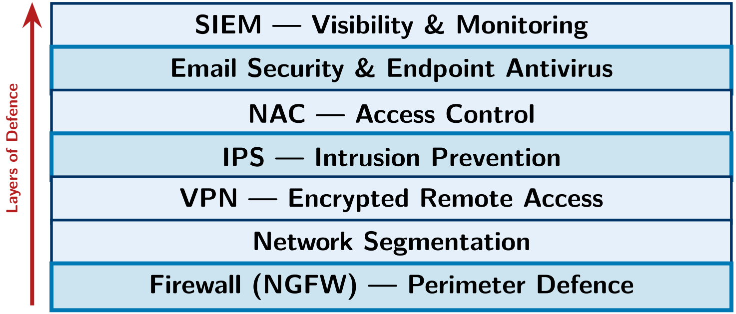
SIEM

- Aggregates logs from **all security tools**
- Real-time **threat detection** and analysis
- Helps security teams **respond** quickly

Network Security — Key Components



Security Technology Stack — Big Picture



Summary

What We Covered

1. Networks: devices, packets, protocols
2. OSI model: 7 layers, focus on Layer 3
3. Network security: definition & CIA Triad
4. Why it matters: data breaches & cost
5. Security controls: physical, technical, administrative

Key Takeaways

- **No single tool** is enough — use layers
- **Zero Trust** is the modern gold standard
- **People & policies** matter as much as tech
- The threat landscape is **always evolving**
- Network security $\subset$ Cybersecurity

Thank You

Questions & Discussion

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